New Frontiers: Regulating Learning in CSCL

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New Frontiers: Regulating Learning in CSCL

Regulation is a Neglected Area in CSCL Research

Two decades of research on computer-supported collaborative learning evidences positive learning effects in two main arenas. First, CSCL technologies have been found to successfully trigger changes in group dynamics by enriching learning interactions and creating opportunities for sharing and constructing knowledge among participants (Ludvigsen, Lund, Rasmussen & Säljö, 2011). Second, CSCL technologies have been associated with changes in both individual and collective collaboration outcomes (Salomon, Perkins & Globerson, 1991). Early research in computer supported collaborative learning (CSCL) drew heavily from collaborative learning and educational technology fields of study. Findings indicated that collaborative learning requires co-construction of a shared understanding (Roschelle & Teasley, 1995), and that productive social interactions can be produced by careful design of CSCL environments (Koschmann, 1996).

Despite extensive research in CSCL over the last decade, there is relatively little research about how groups (and individuals in groups) can be supported to engage in, sustain, and productively regulate collaborative processes. Variations in the quality of collaborative learning processes and outcomes have been consistently documented. Some have even argued that collaborative working procedures and learning processes in CSCL environments may be more regressive than progressive (Veermans & Järvelä 2004; Winters & Alexander, 2010). However, an overemphasis on developing and testing the functionality and usability of technology-based tools for sharing information has diverted attention from the importance of researching how tools can be used to support more effective and productive collaboration. The potential role of CSCL tools for supporting the planning, monitoring, and regulation of collaborative learning processes has been virtually ignored. Gress and Hadwin (2010) identified three significant shortcomings of CSCL research to date. First, support typically refers to developing and testing tools for communicating or sharing, rather than tools for guiding and supporting collaborative processes. Second, research targets knowledge construction processes and products without attending to the processes by which multiple team members join forces to direct and regulate the ways they work together as a team. Finally, CSCL tasks typically require peers to work together (often in dyads) toward individual mastery, rather than creating interdependence or a need for shared learning to occur (Kirschner, 2002).

The purpose of this article is threefold. First, we draw on theory to establish the importance of regulation processes and types of regulation for successful collaboration. This section carefully defines core concepts related to the regulation of learning, distinguishing regulated learning from other often confused constructs in the collaborative learning and CSCL literatures. Second, we illuminate two streams of seemingly diverse research that lay an important foundation for supporting and researching regulation in CSCL contexts establishing that: (a) Web-based pedagogical tools used to successfully support regulation in individual learning contexts can be leveraged for collaborative task contexts; and (b) Computer based tools for supporting collaborative knowledge construction can be leveraged for supporting regulatory processes. Finally, we draw on emerging research to demonstrate how regulation can be supported and explored in CSCL environments. The article concludes by charting a course for future CSCL research focused on supporting regulated learning in collaborative task contexts.

What is regulated learning and why is it important for CSCL?

Our approach to researching regulated learning in CSCL is grounded in early conceptions of self-regulated learning. Self-regulated learning (SRL) refers to the strategic control of thoughts, actions, and motivations to achieve personal goals and adaptively respond to environmental demands (Zimmerman, 2008). From this perspective, regulated learning involves (a) intentionally negotiating task goals and standards to guide work, (b) strategically adopting and adapting tools and strategies to optimize task performance and learning, (c) monitoring progress and intervening if results deviate from plans, and (d) persisting and adapting in the face of challenges (Hadwin & Winne, in press; Schunk & Zimmerman, 1994; Winne & Hadwin, 1998; Zimmerman, 1989). Importantly, SRL theory extends conceptions of *learning* beyond cognitive processes and outcomes, acknowledging the interactive roles of motivation, emotion, metacognition, and strategic behavior in successful learning. There is much research evidence that self-regulated learners are active participants who effectively control their own learning experiences in many ways, including organizing and rehearsing information to be learned, and holding positive beliefs about their capabilities, the value of learning, and the factors that influence learning (e.g. Schunk & Zimmerman, 2008).

Four assumptions of SRL are essential for understanding how regulatory processes influence successful collaboration. First, ***regulated learning is intentional and goal directed*** since goals are interpretations of academic tasks and standards for assessing learning. Goals commit learners to a particular standard or outcome that can be used for self-evaluation. Goals vary in specificity, proximity, hierarchical nature, congruence, difficulty, and process/product orientation (cf. Locke & Latham, 2002; Zimmerman, 2008), but goals are fundamental for regulatory proficiency and success. Knowing something about goals informs researchers about learner direction, motivation, and intent.

Second, ***regulated learning is metacognitive***. The role of metacognitive planning, monitoring, and control processes is central to theories of regulation, and self-regulation in particular. Monitoring refers to comparing a current state with a desired state (goal standard) and making a judgment (evaluation) about goal attainment. Monitoring and evaluating fuel regulated learning. When learners perceive a discrepancy between where they are (individually or collectively) and where they aim to be, an opportunity arises to strategically change thinking, feelings, or actions. Theory and research about regulated learning must explicitly attend to monitoring and control processes such as: activating self/group, task, and strategy knowledge, planning, monitoring, evaluating, or strategically adapting engagement. If metacognitive processes are not measured/observed or systematically analyzed, the research is not about regulation of learning.

Third, ***learning involves regulating behavior, cognition, and/or motivation/emotions***. Research about regulated learning is not about domain knowledge construction. Rather it is about monitoring and controlling thinking, beliefs and strategies to reach a goal. By regulating cognition, motivation and behavior, learners change environmental contexts, themselves, and their groups. For example, if students' personal goals are learning or mastery oriented, the function of regulating emotions is to monitor the conditions necessary for sustained motivation and cognitive engagement towards achievement.

Fourth, ***regulated learning is social***. Whether one views regulation as influenced by environmental context, appropriated through participation, or situated in social activity systems, regulation is social. To understand regulation, one needs to know something about social surround and/or interplay (Schunk & Zimmerman, 1997; Volet, Summers, & Thurman, 2009).

Why is the study of regulation important in the field of CSCL?

Regulated learning is the quintessential skill in collaborative learning. Working together means co-constructing shared task representations, shared goals, and shared strategies. It also means *regulating learning* through shared metacognitive monitoring and control of motivation, cognition, and behavior (Hadwin, Järvelä & Miller, 2010). Without shared task representations and shared goals, collaborative work may become de-railed or less satisfying for learners resulting in less effective, efficient and/or enjoyable learning.

In collaborative learning settings, students' productive engagement in collaborative interactions is viewed as facilitated by the group's coordinated engagement in the shared task space (Roschelle & Teasley, 1995). Achieving coordination is not an easy process, since each group member is a self-regulating agent with unique goals, cognitions, and emotions, which can contribute challenges to motivation in social interactive contexts (Järvelä, Volet, & Järvenoja, 2010). Groups face multiple types of challenges that interfere with the social process of learning and task completion (Blumenfeld, Marx, Soloway, & Krajcik, 1996; Pauli et al, 2007; Webb & Palincsar, 1996). Groups and individuals also face challenges generated by the cognitive processes required in collaborative learning, such as those involved in creating a common ground in shared problem solving (Mäkitalo, Häkkinen, Järvelä, & Leinonen, 2002), negotiating multiple perspectives and handling complex concepts (Feltovich, Spiro, Coulson, & Feltovich, 1996).

We argue that that achieving success in CSCL depends upon: (a) the self-regulatory skills and strategies individuals bring to the group (self-regulation), (b) transitional support provided to one another to facilitate self-regulatory competence within the group (co-regulation), and (c) shared or collective regulation of learning such as metacommunicative awareness, shared motivation regulation and successful coordination of strategies (shared regulation).

Important distinctions between regulated learning and knowledge co-construction

Two cornerstones of successful collaborative learning are (a) shared knowledge construction, and (b) productive collaborative interactions. It is widely accepted that *co-constructing knowledge* involves sharing, transforming, and integrating knowledge through conversational interactions (Resnick, Levine & Teasley, 1991) and sociocognitive interactive dynamics (King, 1998; Salonen, Vauras & Eflikdes, 2005). *Productive collaborative interactions* promote individual cognitive activity which contributes to deep learning (Bereiter, 2002). While knowledge co-construction refers to domain specific and task related aspects, regulation of learning covers sociocognitive and team related aspects (Fransen, Weinberger & Kirschner, this issue).

Contemporary CSCL studies have begun to emphasize the importance of collaborative activities and some models of collaborative learning have acknowledged components that share common ground with regulated learning. For example, processes such as planning, monitoring and evaluating processes and joint understanding (e.g. Hmelo-Silver & Barrows, 2008) or asking questions and requesting explanations (e.g. Scardamalia & Bereiter, 2007) may prompt partners to engage in self-regulatory activities. However, for the most part, these processes are used to prompt collaborative knowledge construction, with little attention devoted to other facets of regulation such as motivation, emotion, strategy use, goals and task perceptions.

CSCL researchers are interested in sociocognitive interaction and they also introduce concepts that partly intersect regulatory constructs. For example, Suthers' (2006) concept of intersubjective learning specifies the process of meaning making in social interactions. Intersubjective learning goes beyond a knowledge sharing conception of collaborative learning in two ways: it can be about sharing interpretations as well as knowledge, and these interpretations can be jointly created through interaction, in addition to being formed by individuals before they are offered to the group. Collective knowledge building (Zhan, Scardamalia, Reeve & Messina, 2009) postulates epistemic agency as a key principle examining how participants take control to reflect on how the community is advancing. Both intersubjective learning and collective knowledge building primarily target cognitive processes leading to shared cognition or the co-construction of domain knowledge. They do not specify the regulatory processes engaged to support co-construction of domain knowledge. In this paper we intentionally distinguish between the domain knowledge that is co-constructed (shared cognition) and the individual and shared strategic processes that lead to shared domain knowledge and knowledge co-construction.

We propose that successfully collaborating in CSCL contexts requires targeted support for promoting or guiding: (a) individual self-regulatory skills and strategies, (b) peer support and facilitation of self-regulatory competence within the group, and (c) shared or collective regulation of learning. From this perspective, collecting data about knowledge, distributed knowledge, or co-constructed meaning alone is not sufficient to understand or make empirical claims about regulated learning. However, it provides essential information about the outcomes of regulatory processes. We posit that research about the regulation of learning needs to extend beyond the regulation of knowledge construction to consider such as the processes by which: (a) knowledge of tasks, strategies, goals and group are regulated during collaborative work, (b) knowledge of self, collective and each other's beliefs, feelings and motivations are regulated, and (c) behaviors and actions with a group are regulated to influence collaborative outcomes.

What distinguishes shared knowledge construction from shared regulation is the target information. In shared knowledge construction, knowledge is constructed about the domain. Strategies are about the domain knowledge construction such as finding the right answer or solving the problem. In shared regulation, metacognitive, meta-motivational, and meta-emotional knowledge is constructed about the collaborative learning processes, such as negotiating and aligning representations of task requirements and goals. Sharing in the process of monitoring and evaluating progress, and constructing strategies that might help a group to work more effectively and efficiently together. A second distinction is that regulated learning focuses on learning and collaboration processes, rather than processes and outcomes associated with domain task completion.

Three types of regulated learning contributing to collaborative success

When individual learners work collaboratively, we argue that at least three types of regulated learning come into play (see Figure 1). In all three types of regulation, the term *learning* is used liberally extending beyond cognitive knowledge construction to actions (behaviors), beliefs (motivation and effect) and cognitive processes or strategies (cognition). Successful collaboration requires: (1) Each group member to take responsibility for regulating his or her learning (self-regulated learning). (2) Each member to support fellow group members to successfully regulate their learning (co-regulated learning), and (3) The group to come together to collectively regulate learning processes in a synchronized and productive manner (shared-regulation of learning). We discuss each type of regulation in more detail below.

Insert Figure 1. about here

Grounded in social cognitive theory, *self-regulation* is guided by environmental conditions that promote individuals to adopt, develop, and refine strategies, monitor, evaluate, set goals, plan, adopt and change belief processes (Schunk & Zimmermann, 2008). These self-regulatory processes operate when students work on solo tasks and when they work together in complex collaborative tasks. From this perspective, successful team work requires each group member to regulate his/her own cognitive processes, actions and beliefs (Winne, Hadwin, & Perry, 2012). For example, in Table 1, each group member takes responsibility for setting goals/ standards for his/her own contributions to the team. Individuals regulate their own actions and beliefs toward the group task. Table 1 details the way self-regulation occurs in collaboration with respect to three main questions: Whose goals? Who regulates? What is regulated? It also provides a couple of examples of dialogue or reflections that illustrate each form of regulated learning. We argue productively self-regulating oneself in a group task is a condition for successful collaborative learning, but SRL alone is not sufficient for collaborative learning to occur. Even when student dyads appear to be working together to share information, learning gains may be imbalanced between the individuals, suggesting students are self-regulating their own learning but not regulating together (Winters & Alexander, 2010).

Co-regulation occurs when individuals' regulatory activities are guided, supported, shaped or constrained by and with others. Interdependence exists when goal and strategy knowledge and processes are reciprocally shaped by one another. Co-regulation requires team members to be aware of one another's goals and progress and to consider those in relation to the shared task. For example, in Table 1, students support each other's regulation toward a group goal by prompting one another to regulate or to recognize discrepancies between their own actions, beliefs or perceptions and goals or standards the group has negotiated. In the context of collaborative work, engaging in co-regulation involves developing awareness of one another's regulatory processes including affordances and constraints for regulation that emerge at the intersect of people, CSCL tools, tasks, and contexts. Regulatory support may be distributed amongst group members, but the outcome of co-regulation is that each individual's regulatory activity is supported and shaped by interactions with one another (Volet, Summers & Thurman, 2009).

Co-regulation is a central process in collaborative learning tasks. In addition to monitoring and regulating one's own processes, successful collaboration requires individuals to actively monitor and regulate each other through questioning, prompting and restating. It involves also mutual performance monitoring for keeping track of each other's work (Fransen et al., this issue). Much like self-regulation, we argue that co-regulation is necessary for successful collaborative learning to occur (Winne, Hadwin, & Perry, in press). However successful collaboration requires more than self-regulated and co-regulated learning.

Shared regulation occurs when groups regulate as a collective such as when they construct shared task perceptions or shared goals. In this case, goals and standards are co-constructed, and regulation is distributed and shared with multiple ideas and perspectives being weighed and negotiated until consensus is met. For example, in Table 1, team members share individual perspectives collectively constructing a shared task perceptions. During this process new ideas may emerge and divergent ideas may be shared, discussed and either included or excluded. When groups co-construct plans or align monitoring perceptions to establish a shared perception or evaluation of progress, they are engaged in shared regulation. Socially shared regulation of learning involves the construction and maintenance of interdependent or collectively shared regulatory processes, beliefs, and knowledge (e.g., strategies, monitoring, evaluation, goal setting, motivation, metacognitive decision making) orchestrated in the service of a co-constructed or shared outcome (Hadwin, Järvelä & Miller, 2011).

Insert Table 1 about here

Findings to date indicate that different aspects of shared regulation emerge when students work in collaborative learning groups and make consistent efforts to regulate learning and engagement. Students, for example, shape their use of motivation regulation strategies to fit the specific situational group challenges and activate new strategies for motivation, such as "social reinforcing" (Järvelä, Järvenoja & Veermans, 2008). Also findings in emotion regulation show that in socio-emotionally challenging situations emotional balance can be restored if regulation of emotion is constructed in interaction between the group members (Järvenoja & Järvelä, 2009).

Research also indicates that students do engage metacognitive processes in collaborative problem solving contexts (Hurme & Järvelä, 2005) and that metacognitive knowledge and metacognitive skills are distributed amongst individuals in problem solving groups (Hurme, Palonen, & Järvelä, 2006). However, in group tasks different types of metacognitive regulation emerge including: metacognition becoming shared, metacognition becoming visible but not shared, and individuals sharing metacognition by attempting to regulate joint activity (Hurme, Merenluoto, & Järvelä, 2009).

In sum, findings illuminate socially shared regulation as a complex and extraordinary process that plays a central role in collaborative success (Kempler Rogat & Linnenbrink-Garcia, 2011). Findings indicate that students can regulate emotions, motivation, cognition and metacognition together as well as individually (Järvelä & Järvenoja, 2011; Hurme, et al. 2009; Volet et al., 2009) demonstrating that understanding regulation requires attention to both individual- and group-level data (Salomon & Perkins, 1998).

Promoting Regulation in CSCL

Increasingly, scholars recognize the importance of regulated learning in technology enhanced learning environments and CSCL environments (Dabbagh & Kitsantas, 2004; Hartley &

Bendixen, 2001; Schunk & Zimmerman, 1998). Students must exercise a high degree of self-regulatory competence to accomplish their learning goals, whereas in traditional face-to-face classroom settings, the instructor exercises significant control over the learning process and is able to co-regulate learning by monitoring student attention and progress closely. The effective use of SRL strategies is essential in flexible learning situations due to the high degree of student autonomy.

The field of CSCL has been challenged by its interdisciplinary nature spanning the learning sciences, educational technology, educational psychology, and computer science. One outcome is that "support" has been equated with the provision of tools and technologies for enabling collaborative discussion and exchange. A limitation of this emphasis is that it ignores the fact that providing mechanisms for collaboration does not insure teams know how to leverage those tools to optimize collaboration.

We argue that advancing the CSCL field requires attention to the roles technology can play in supporting teams to collaborate more productively by optimizing self-regulated learning, co-regulated learning and shared regulation processes. Despite the paucity of research examining tools for supporting regulation in CSCL contexts, two seemingly disconnected lines of research offer promise for the field. First, technologies in the form of computer-based pedagogical tools (CBPT) have been successfully implemented to enhance regulated learning processes in solo contexts. Second, supports for cognitive and functional aspects of collaboration and knowledge construction have been successfully implemented to structure the collaborative interactions.

CBPT can be used to support SRL in solo tasks: Implications for CSCL

Recent research about computer-based pedagogical tools has clearly established that these technologies can be leveraged to support self-regulated learning. While work in this area focuses predominantly on solo rather than collaborative computer supported learning environments, it sets the stage for designing and testing the efficacy of technologies to support regulated learning in collaborative task contexts.

The emergence of CBPTs embedded in various technology enhanced learning environments have shown great promise for enriching web-based instruction by providing tailored support for self-regulated learning processes. This especially to support “I” perspective”, such as “my goals and my plans”, but also temporary support for “you perspective” supporting, for example, each others task perceptions in co-regulation (See Figure 1.). CBPTs are not designed for collaboration, but may be considered early prototypes for designing tools to support regulation and offer direction for embedding regulation supports in CSCL contexts. CBPTs have proven to be particularly effective for supporting regulation in college classrooms where students are expected to function independently with limited student-teacher interaction. A number of studies have employed some type of SRL scaffolding through: (a) peers (De Jong, Köllöffel, van der Meijden, Staarman & Janssen, 2005; Winters and Azevedo 2005); (b) tools embedded in the environment (Narciss, Proske & Koerndle, 2007) or (c) tutors (Azevedo, Cromley & Seibert, 2004).

CBPTs usually provide four types of scaffolds: conceptual, metacognitive, procedural and strategic scaffolds (Fischer, Kollar, Stegmann & Wecker, this issue; Hannafin, Land & Oliver, 1999). Conceptual scaffolds include hints and prompts designed to guide what knowledge to consider during problem solving and understanding complex topics (Liu & Hmelo-Silver, 2009). Metacognitive scaffolds include, for example, agents that help students with specific task-related tasks. These agents help students self-regulate the underlying processes associated with managing learning (White, Shimoda, & Frederiksen, 2000). Procedural scaffolds assist students with learning how to use resources or perform tasks (Azevedo, Verona, & Cromley, 2001). Strategic scaffolds make students aware of different techniques for solving problems and expose them to peer or expert solution paths (Lajoie, Guerrera, Munsie, & Lavigne, 2001).

Research examining CBPTs have investigated the complex interplay between learner characteristics (e.g. prior knowledge), elements of technology tools, and mediated self-regulated learning processes (e.g. planning, strategy use, monitoring) used by students during hypermedia learning. Findings indicate CBPTs can prompt students to set appropriate goals, self-monitor, self-evaluate and ask for help as needed (Dabbagh & Kitsantas, 2005). WBPTs can also be used to scaffold the acquisition of metacognitive skills and strategic learning such as knowledge of plans and goals (Azevedo, Cromley, Winters, Moos & Greene, 2005), and engage students in complex and thought-revealing learning tasks such as self-reflection, problem-solving, and social negotiation. These activities increase self-awareness and self-observation, which are key attributes of metacognition and self-regulated learning.

Typically CBPT research involves: (a) solo tasks because the desired products are individual understanding or mental models, and (b) technology or human tutors used to support self-regulated or other-regulated learning. In most cases, scaffolding has been tuned to individual processes and knowledge outcomes rather than prompting groups' processes, monitoring, awareness, or strategic adaptation of learning process.

Despite the emphasis on solo work contexts and tasks, research about CBPTs offers promise for informing the design, implementation and evaluation of CSCL supports targeting regulation. For example, De Jong et al. (2005) extended CBPT research to investigate students' engagement and regulatory activities in 3D and CSCL environments during collaborative problem solving. Analysis of students' interaction indicated that basic self-regulation processes, such as monitoring, directing and testing were less frequent than co-regulatory activities, such as grounding and making common agreement. Negotiation of shared goals is part of the interactive process of grounding. Building and maintaining common ground means individuals construct shared understanding, knowledge and beliefs (Clark & Schaefer, 1989). Findings also revealed aspects of co-regulation such as confirming that their partners understood what they had been said (shared monitoring of the task progress).

CSCL tools can be used to support knowledge construction: Implications for supporting regulated learning

A second line of research offering promise for supporting regulation in CSCL contexts involves the successful use of various CSCL supports for cognitive and functional aspects of collaboration. While this body of research doesn't focus on supporting regulation per se, it does introduce tools and support mechanisms that can be leveraged for the purpose of supporting regulation in CSCL contexts.

CSCL support is interpreted broadly in the CSCL literature. For this paper, we focused exclusively on supports internal to the CSCL task environment (Morris, Hadwin, Gress, Miller, Fior, Church, & Winne, 2010), deliberately excluding support provided external to the CSCL task environment in the form of tools, resources, and person support. Internal supports are anchored in the immediate CSCL task environment and are used to carry out some of the collaborative work including tools that: (1) *enable* collaboration such as whiteboards, wiki spaces, chat tools, and file sharing tools, and (2) *support*, guide or make visible the collaborative process. A decade long explosion in technologies and environments for enabling collaboration, has lead to an overemphasis on usability and an under-emphasis on the ways technologies might be leveraged to guide or support collaboration (Morris et al., 2010). Simply providing usable and accessible technologies to enable collaboration has not proven to be fruitful for leveraging collaborative potential in distributed groups.

While technologies that support collaborative processes are relatively new on the CSCL scene, they offer potential for being leveraged to support the collaborative regulation of learning: CSCL supports include: (a) structuring supports, (b) mirroring or visualization supports, (c) metacognitive awareness tools, and (d) guiding tools or systems (Soller, et al., 2005).

Structuring supports strive to enhance collaborative processes and outcomes by structuring, scripting and prompting collaboration. They have been used to support functions and procedures for collaborating (functional supports) as well as to guide conceptual understanding and processing (cognitive supports). Structuring supports include role assignment, scripts for collaborating and prompts for taking on particular collaborative roles or functions.

Roles are prescribed functions that guide individual contributions to collaborative work. Early work focused on functional roles for task completion functions such as data collector role (Strijbos, Martens, Jochems, & Broers, 2004), or recorder/notetaker, materials manager, and editor roles (Slavin, 1995). In comparison, cognitive roles focus on supporting thinking, processing and understanding in the task (O'Donnell, Hmelo-Silver, & Erkens, 2005). *Scripts* consist of instructions for carrying out collaboration. They are akin to recipes for acting out roles or for coordinating roles. Social scripts describe how to structure discourse and interaction. Epistemic scripts describe cognitive processes and strategies to be used (c.f. Weinberger, Ertl, Fischer, & Mandl, 2005). *Prompts* are reminders, usually in the form of sentence openers or question stems, to help individuals carry out prescribed roles in the way they were scripted.

To date, roles, scripts and prompts have been used to structure research social, epistemic, and cognitive/conceptual engagement (Weinberger et al., 2005). Strijbos et al. (2004) found students assigned functional roles (project planner, communicator, editor, etc.) didn't perform differently on the collaborative task than students in a no roles condition, but did indicate higher perceived group efficiency and engage in a higher frequency of task-content focused statements. A limitation of work in this area is that roles, scripts and prompts have not been leveraged or researched as supports for self-, co-, and shared regulation processes (e.g., planning, strategizing, monitoring). Hadwin et al. (2010) proposed ways to script and support the regulation of collaboration using these types of roles and prompts. For example, a chat tool might provide sentence starters and questions to guide students to discuss and construct share task perceptions (e.g., Why are we being asked to do this assignment), consensus about goals and standards (e.g., What is our goal for this task? What standards are appropriate for us in this task?), strategies for collaborative task execution, and ongoing monitoring and evaluation of process and product (How we doing? What are our biggest challenges right now? Are we all concerned about…). Combining these shared regulation prompts with roles and scripts that guide collaborators to take responsibility for, and systematically work through each regulation phase represents a promising line of inquiry because it shifts the target of support from functional coordination or the knowledge co-construction, to successfully planning, engaging, reflecting upon and adapting collaborative processes to optimize emerging potential of a team. Support might target improved self-regulation in a collaborative task, improved co-regulation in a collaborative task and/or improved shared regulation in a collaborative task (See Figure 1.).

Mirroring and Metacognitive Tools. The second type of CSCL support comes in the form of mirroring tools that collect, aggregate and reflect data back to the users about individual and collective interaction and engagement (Soller et al., 2005). The goal of this type of support is to make collaborators aware of individual or collective actions thereby inviting collaborators to monitor, evaluate, interpret and act on that information themselves. A third type CSCL support builds on mirroring by providing visualization and aggregate representations of collaborative work and social activity (e.g. Phielix, Prins, Kirschner, Erkens, & Jaspers, 2011), but these tools go a step further to align those representations with expert models or information about standards for optimal collaboration.

Mirroring support tools target: (a) group awareness such as knowing who is online, where collaborators are in the elearning system, what they are looking at and what they are doing (e.g., Kreijns, Kirschner, & Jochems, 2002; Leinonen, Järvelä, & Häkkinen, 2005), as well as (b) social awareness such as knowing team members perceptions, ratings, or knowledge (Buder & Bodemer, 2008). When information is presented in such a way to promote comparisons between team members or across teams, the tools become metacognitive tools. For example, Buder and Bodemer described an awareness tool where each team member's statement of opinion was accompanied by ratings of novelty and agreement. A graph with level of novelty on the y-axis and agreement of the x-axis displayed the position (intersect) of each team member's ratings. In this way teams could see if they are well aligned or poorly aligned based on the proximity of their points. Below the graph each opinion statement and rating was provided below the graph making it easy to compare position to the actual ideas and ratings that were expressed by teach team member. Buder and Bodemer (2008) found students who were given an augmented awareness tool demonstrated higher performance in terms of the group decision on to a problem case as well as individual correctness compared to a group in an unsupported control condition. Consistent with the social psychology literature, they also found students who did not have the awareness support tended to defer to the majority in their decision making.

Notification and awareness tools have also been used to synchonize task-oriented collaborative activity (Carroll, Neale, Isenhour, Rosson, & McCricklard, 2003) including: (a) Social awareness regarding presence and motivational states of collaborators, timing, frequency and intensity of activity and communication, (b) Action awareness regarding timing and frequency of interaction with a shared source, and (c) Activity awareness regarding changes to shared plans or evaluations, modifications to project roles, task dependencies based on roles, etc. By dislaying critical situational, task, tool, and group information this system supported: (a) mirroring, aggregating and synthesizing information to invite monitoring and regulation, and (b) metacognitive tools for comparing across team members or time.

Mirroring and metacognitive tools have tremendous potential for supporting regulation in collaborative tasks. Visualizing can be used to represent each group member's ratings of the accuracy of task criteria they have identified as well as their certainty that those are the criteria. Awareness about diversity in responses with the team, creates an opportunity for the team to evaluate that task perceptions may not be complete and to do something about that (regulate). Similarly, visualizing each team member's perception of challenges encountered in the task, invites team members to negotiate and construct strategies to address challenges. In these examples, regulation is the target rather than task completion, domain knowledge acquisition, or distribution of work. To date, mirroring tools have not been used to support or research the self-regulated, co-regulated, or shared regulation of learning. Research has been limited to small scale lab studies. Phielix et al.'s (2011) study of high school learners is one exception because they used mirroring tools in collaborative writing tasks to assess group members', themselves, and their fellow group members social and cognitive behavior. Mirroring tools were also used to stimulate group members to reflect on past, present and future group functioning, encouraging goal setting and planning for improved social and cognitive performance.

Guiding Systems

The most sophisticated type of CSCL support is guiding systems that act on discrepancies between data collected (mirroring data) and standards for optimal collaborative functioning to guide or externally regulate collaborative processes and communication (Soller et al., 2005). Guiding systems use complex computational algorithms to interpret data and coach collaboration. Guiding systems also present potential challenges for supporting regulation. They must be implemented to progressively move the locus of control back to collaborators, or they risk making learners dependent upon other-regulation for collaborative success. The goal should always be to guide students to take active control and responsibility for regulatory processes in collaboration.

For example, Figure 2 (adapted from Miller & Hadwin, 2012) shows a visualization of individual perceptions of the purpose of a group task that has been assigned in a course. Each grey idea is something an individual identified themselves but did not share or discuss with the group. Responses on the right display the task purpose as negotiated by a group. The table below is a visualization of individual's post task analysis indicating that each team member thought they had come to consensus about the task purpose. Together these visualizations promote awareness for the group that they may not be as well aligned with each other as they think. A guiding tool would apply an algorithm to guide or prompt students to correct this deficiency in their negotiation of task. For example, it might display all the individual ideas that were not discussed or shared and ask the team to actively reject or accept each of them.

Insert Figure 2. about here

New Frontiers: Researching shared regulation in CSCL

In our current research together with colleagues we have focused on promoting and sustaining socially shared regulatory processes in CSCL, such as collaborative goal setting, task understanding, cognitive and motivational regulation and adaptation, beyond the introduction of collaborative software tools (Hadwin, Oshige, Gress, & Winne 2010; Järvelä & Järvenoja, 2011). We strive to support collaborative process through innovative collaborative tools and technological scaffolds designed, for example, in the context of the Learning Kit research project (Winne, Nesbit, Kumar, Hadwin, Lajoie, & Azevedo, 2006).

Research about shared regulation of learning focuses on what is shared or co-dependent in terms of (a) regulated learning knowledge, beliefs, and processes and (b) co-constructed planning, monitoring, evaluating, and strategy regulation processes such as: shared task perceptions, shared goals, shared plans, shared monitoring and evaluation, and shared strategies (Hadwin, Järvelä & Miller, 2011). CSCL technologies offer benefits for promoting socially shared regulation of learning by helping collaborators search for and organize information solo and collaboratively, and prompting learners to metacognitively consider features of their work across levels of self, co- and shared regulation (Hadwin et al., 2010). This is to prompt to students' "my", "your" and "our" perspectives to regulate learning as described in Figure 1. Today's technologies can collect log data about almost every observable facet of these events unobtrusively in formats that are ready for further, sometimes automated analysis and, thus, provide evidence for shared monitoring and evaluation (Winne et al., 2006). Recent mobile, handheld supported collaborative learning can structure and regulate individual and collaborative learning strategies, for example, scaffold externalization of knowledge representations in individual and collaborative levels in order to maintain collectively shared regulatory processes

(Järvelä, Näykki, Laru, & Luokkanen, 2007) or bridge learning activities in formal (e.g. classroom activities) and informal contexts (e.g. museums) for shared collaborative learning activities (Kurti, Spikol, & Milrad, 2008).

For example, Laru, Näykki and Järvelä (2010) used multiple social software tools (e.g. wikis), smartphones and face-to-face activities to structure collaborative learning among teacher education students' educational technology course. The tools (e.g. taking pictures by smartphones and sharing them in wikis) supported interaction between individuals and their collective actions and helped to share task perceptions, plans and monitoring, resulting in better learning amongst students.

Our next examples draw from our research collaborations regarding regulation in collaborative contexts. The examples reflect our respective interests in emotion regulation, and shared metacognitive planning. However, future research will examine other aspects of shared regulation including cognitive and motivational processes. In our study Järvelä et al. (2012) used a set of CSCL embedded tools to facilitate self- and shared-regulation among 18 graduate students enrolled in a CSCL course. The course included three learning cycles each consisting of a: (a) one-day face-to-face meeting, (b) one-week on-line solo studying period, and (c) one-week virtual collaborative task period. Online tools were designed to support and sustain productive regulation of learning at individual and group levels. Supports were used in different phases of self-regulated learning and they were embedded into nStudy learning environment (Winne et al., 2006).

Students' engagement in collaborative learning processes was supported by the three tools: (a) Emotion awareness tool, (b) Planning and reflection tool, and (c) Adaptive instrument for regulation of emotions. Every time the students logged in to the nStudy learning environment they were advised to fill in an “Emotion Awareness tool” (Järvenoja, Järvelä & Malmberg, 2011) as prompting students to respond to three questions in the moment: 1) How are you feeling right now, 2) Why? and 3) How do you aim to control your emotional state? This tool was targeted the forethought phase of self-regulation by prompting awareness of the emotional state when starting studying. It also scaffolded executive phase, when emotional state calls for emotion and motivation regulation. In other words, the tool was designed to help students to self-regulate and monitor motivation and emotion regulation and to identify goals or standards they could use to monitor and regulate emotions during collaboration.

A second tool provided in the CSCL environment was a “Collaborative Planning and Reflection tool” that scripted planning by prompting students to: (a) negotiate and co-construct task perceptions and goals for a collaborative task (planning), (b) reflect on the challenges encountered for that task, and (c) construct strategies for dealing with those challenges should they occur in a future task (Hadwin, Malmberg, Järvelä & Järvenoja, 2012). This planning and reflection cycle was designed to promote shared regulation on this collaborative task and to feed forward to future collaborations in the course. Researchers compared, shared responses to individual regulatory statements regarding task perceptions and goals for the same collaborative task. Findings of a cross-case comparison indicated that groups successfully negotiated accurate perceptions of the explicit features of tasks they were assigned. However, groups did not successfully negotiate shared perceptions of the implicit aspects of the tasks that actually required some inference and discussion (such as why are we doing this, or what course concepts should we be using). Furthermore, an analysis of shared goal statements indicated that teams did not reflect or build upon goals individuals held for the same task.

In a second study examining shared planning, Miller and Hadwin (2012) most group members reported having high consensus about shared plans (task perceptions and goals). However analysis of their task planning negotiations indicated that none of the groups systematically solicited or discussed individual perceptions of task purpose or goals. The higher performers converged on common ideas, neglecting to share or discuss divergent ideas held by single members of the group, even when those task perceptions and goals were well aligned with the collaborative task.

Together, these findings point to the importance of extending prompts and scaffolds with mirroring and metacognitive tools that serve to increase awareness and calibration between individual and collective goals and plans. In this case, teams may benefit from visual cues showing how each team member`s individual goals align (or do not align) with the team goal. For example, viewing a graphic such as the one shown in Figure 2, creates an opportunity for teams to consider the degree to which they capitalized on each other's knowledge to collectively plan for the task. Viewed during collaboration, these types of visualization may prompt deeper planning and negotiation.The third tool implemented in the CSCL design was the "Adaptive Instrument for Regulation of Emotions" (AIRE) (Järvenoja, Volet & Järvelä, 2012) that guided students to reflect upon socio-emotional challenges encountered during collaboration. The AIRE prompted them to: (a) identify challenges they experienced during collaboration, and (b) reflect on the ways emotional experiences affected collaboration. Additionally, the AIRE responses were used to generate feedback about students' own responses and their responses in comparison to other team members between collaborative tasks. This feedback created opportunities for group members to calibrate and monitor standards and evaluate the progress of collaboration. For example, some of the group members reported that differences in personal priorities created a great challenge and required regulation across the course, while some others did not see it as a challenge at all. By using AIRE, the group members became aware of each others' individual interpretations of the collaboration. By receiving group level feedback on their AIRE responses, they also formed an understanding how they experienced the collaboration as a group and hence had an opportunity to calibrate their shared regulation to fit the needs of the group.

Results from this research, to date, indicate that the students can be supported to develop shared regulation. Strong shared regulation within the group led to use of deep level strategies instead of routine level regulation strategies (Järvelä et al., 2012). Importantly, to optimize opportunities for shared regulation, our findings suggest that both individuals and groups need to engage regulatory processes and strategies. However, planning, monitoring and reflecting alone are not sufficient for shared regulation. Teams need to be supported to increase awareness of how collective and individual perceptions, strategies, beliefs and processes align (Gress & Hadwin, 2010). Having situational awareness of individual and collective perceptions of and experiences with collaborative learning, invites regulation.

Future directions: Researching self-regulated, co-regulated and shared regulation of learning in CSCL contexts

This paper argues that successful learning in CSCL environments depends in part on developing, applying and adapting regulatory skills and strategies. Merely putting students in groups and providing technologies and tools to collaborate is not sufficient for optimizing learning and co-construction. Left to their own devices, research indicates that students do not effectively regulate either individually or collectively. Current CSCL research examines tools and supports for optimizing knowledge competence, attitudes about learning and collaboration (Van den Bossche, Gijselaers, Segers & Kirschner, 2006), knowledge construction (Zhang, Scardamalia, Lamon, Messina & Reeve, 2007) and team related issues (Saab, 2012). However, there is a paucity of research examining the role regulatory processes or regulatory supports play in determining the quality and efficiency of collaborative task work. In order to advance research in CSCL, we advocate for systematically examining tools, supports, and technologies for promoting regulation, and shared regulation in particular, within CSCL contexts. We posit that optimally functioning groups bring together all three forms of regulatory competence and engagement: team members regulate their own task engagement (self-regulated learning), they transitionally support each other to regulate (co-regulated learning), and they regulate as a collective entity (shared regulation). To advance research about shared regulation, we propose four targets for the CSCL field:

Target 1: Distinguish shared regulation from shared knowledge construction

Recent empirical research has shown that it has been especially challenging to differentiate shared regulation from shared knowledge construction. The concepts of “sharing,” “regulation” and “co-regulation” have been increasingly used in research literature, but without explicit definition or clearly conceptually grounded operationalization (Hadwin et al., 2010). For example, in a review of team learning, Decuyper, Dochy & Van den Bossche (2010) claim that the depth and breadt of sharing determine the quality of team learning. Specifically, they highlight the importance of “team reflexivity”, which they define as “processes of co-constructing, de-constructing and re-constructing shared mental models about current reality, and about team goals and methods (Decuyper et al. 2010, p. 117)”. While we agree that these types of shared knowledge construction processes are central to successful collaboration, we disagree that they are synonymous with shared regulation. Shared regulation specifically targets metacognitive, motivational and strategic knowledge and processes.

Target 2: Increase individual and group awareness of regulatory processes

Regulation implies adaptation or change. In order for learners to regulate during collaborative work, they need to be aware of their own regulatory processes, the regulatory processes of team mates and where the group as a whole stands with respect to regulation. In order to recognize that something is going awry and that the collaborative team needs to realign task perceptions, goals, strategies, or socio-emotional states, the group (collectively and individually) need to develop shared social, action and activity awareness. Social awareness and group awareness tools have been found to be effective in providing information about the group's collective actions and visualizing social interaction (Kirschner & Kreijns, 2003), but more research is needed to investigate how shared awareness tools can be used to increase collective awareness of regulatory activities, such as shared goals, knowledge about the strategies group members have used together or evaluations of collective progress. For example, research designs that aim explicitly to increase students' awareness of their motivation and their role as active regulators of learning can effectively channel new knowledge back into practice, as data are generated. Methods such as AIRE instrument (Järvenoja, Volet & Järvelä, 2012) provided a way to increase both teacher and student awareness of emotional and motivational aspects that are situation-specific. This type of information can be channeled to scaffold learners' regulation processes and support them to become active agents in their own learning.

Target 3: Develop research designs for investigating solo and shared strategic regulation of learning

Self-regulated learning, co-regulated learning and shared regulation of learning evolve together during collaborative tasks. Rather than existing as separate entities they might be thought of as existing along a continuum. In order to research and examine shared regulation, we need to broaden and extend our analytic approaches. We have suggested contextual and task-specific data collection methods to investigate self-, co- and shared regulation so that it can be empirically distinguished (Järvelä, Järvenoja & Näykki, 2012). Data need to be collected and examined over larger episodes of time and across events to contextualize and calibrate amongst individual and collective representations of shared regulatory processes. Examining how individuals and teams respond to task and process challenges may create opportunities to examine individual and collective monitoring, evaluating and adapting of task perceptions, goals, strategies, and motivational states in a shared social space. We posit that identifying episodes of challenge in collaborative learning environments and following how individuals and collectives respond to and resolve those challenges is central for examining regulated learning.

Target 4: Leverage SRL theory and tools to the new frontiers in CSCL research

The field of CSCL research has been approached at least from two perspectives: (a) making collaborative learning more functional by supporting collaborative interactions and team related aspects (Fransen et al., 2010), as well as (b) supporting co-construction of knowledge and other domain specific or task-related aspects (Saab, Van Joolingen & Van Hout-Wolters, 2012). Currently, in the CSCL research many overlapping concepts emerge with various foci. For example, task regulation and team regulation in CSCL can be characterized in various levels of social interaction, but the target of the analysis varies from metacognition to social activity (Saab, 2012). These studies contribute to CSCL research by stressing the impact of regulation in collaboration, but support self-regulation, co-regulation and socially shared regulation in co-construction of shared task representations, goals and strategies in CSCL presents new and exciting directions for CSCL research.

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